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ANALYTICAL REPORT

Planning Benefit-Risk Assessments Using Visualizations

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Abstract
A Benefit Risk Assessment Plan (BRAP) describes the assessments planned to determine whether the benefits of an investigational drug outweigh the risks. The plan can have two sections, one with timelines for aligning resources with decision milestones and the other for pre-specifying assessments for decision milestones. Regulatory guidance recommends a proactive planning process over an ad-hoc process. However, very little has been published about proactive plans themselves. This article works through a hypothetical example visualizing a series of assessments across the drug development lifecycle. Based on a regulatory framework, the planning process starts with assessing the medical condition and current treatment options. These early assessments bring out major considerations in assessing the investigational drug.

Keywords Decision-making · Planning · Uncertainty · Tradeoff · MCDA

Background

Three significant developments in the benefit-risk field were the IMI PROTECT (Pharmacoepidemiological Research on Outcomes of Therapeutics by a European Consortium) initiative [1] for choosing methodologies (2009–2015), the Benefit-Risk Assessment Tool (CIRS-BRAT) [2] for generating value trees and forest plots (2009), and the FDA Benefit-Risk Framework [3] (see Table 1) for reporting and reviewing assessments (2014). At that time, planning was not essential. A forest plot assessment could easily be included in a statistical analysis plan (SAP). Today, a single assessment has grown into a series of assessments spanning the entire drug development lifecycle. The questions and decisions become more complex at each subsequent assessment, requiring different data types and visual displays. These developments necessitate a stand-alone BRAP that supports an explicit, transparent decision-making process that balances benefits and risks while integrating value judgment. Proper planning ensures the right data are collected, right methods are applied, and the right questions are answered in a timely fashion. Supporting this argument, the United States Food and Drug Administration (FDA) in September 2021 released its "Benefit-risk Assessment for New Drug and Biological Products Guidance for Industry" [4], which states, "Generating rigorous evidence to inform FDA's benefit-risk assessment calls for careful planning and should involve prospective interaction with FDA, as well as complete documentation of selection of methodology, data, assumptions, results, and sensitivity analysis of uncertainties." The Council for International Organizations of Medical Sciences (CIOMS) draft guidance "Benefit-risk balance for medicinal products" [5] in 2023 states, "A BR-focused proactive planning process should be considered during selection of study design, study population, dosing strength and frequency, inclusion and exclusion criteria, risk mitigation measures, as well as study endpoints and clinical outcome assessments."

Unfortunately, little has been published on writing a BRAP [6]. The IMI Benefit-Risk Group recommended a plan using a decision framework known as PrOACT-URL (Problems, Objectives, Alternatives, Consequences, Tradeoffs, Uncertainty, Risk attitudes, and Linked decisions), which is an eight-step process rather than a plan [7, 8]. The CIOMS guidance proposes a Benefit-Risk Assessment Document (BRAD) based on the FDA framework, including

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Table 1 FDA Benefit-Risk Framework and Assessment [1]

Dimension	Evidence and Uncertainties	Conclusions and Reasons
Analysis of Condition		
Current Treatment Options		
Benefits		
Risks and Risk Management		
Conclusions Regarding Benefit-Risk		

risk mitigation and conclusions. It is a living document that records results and how assessment decisions were made.

Recommendation

This paper recommends a decision-making process aligned with the FDA framework, whereby each assessment builds off the prior assessment. Given that a plan is a form of communication between stakeholders, we use visualizations to communicate the benefit-risk tradeoffs and how value judgment is integrated with clinical evidence. To make the examples generalizable to any treatment, we use a hypothetical example. We also recommend a BRAP that meets the following criteria:

A BRAP is:

- A living document that can be updated at each decision milestone during the drug development lifecycle
- A description of each planned assessment, assumptions, and data source
- A plan to prospectively address uncertainties impacting the decision of whether the benefits outweigh the risks
- A plan to assess multiple outcomes having different scales and weights
- Documentation of how outcomes, statistical characterization, methods, thresholds, and weights were selected
- Facilitation of stakeholder discussions, including regulators, patients, physicians, payers, and sponsor departments,
- A plan aligned with documents, such as the Target Product Profile (TPP), Clinical Development Plan (CDP), Integrated Statistical Analysis Plan (ISAP), and Program Safety Analysis Plan (PSAP)
- Strictly a planning document and not to be confused with a reporting or review document, which would include the conclusions at the bottom of the FDA framework.

For our test case, we chose a hypothetical chronic condition manifested as either Type A or B, with three levels of severity and impacting patients' physical, social, and mental functioning. Five comorbidities accompany the condition. There are four current treatment options drugs A, B, C, and D. Associated adverse events (AEs) include recurring AEs and a rare but serious AE. Follow-up time was 36 weeks.

Terminology

Discussions among stakeholders will require using common terminology [9] listed in Table 2.

Structure of a BRAP

Our draft BRAP has two main sections, the drug development timeline, and the pre-specification of assessments. The value of the timeline is (1) scheduling resources and data access to meet decision milestones. The value of the pre-specification of assessments is transparency in decision-making and insights gained from assessing current treatments first. The Gantt chart in Figure 1 displays the timelines. It starts with the decision milestones at the top, such as Phase 2A Go-No-Go decisions, End of Phase 2B regulatory meetings, and End of Phase 3 FDA advisory committee meetings [9]. More decision milestones can be added to each development phase. The BRAP needs to be aligned with strategic documents in the middle row, such as the TPP, CDP, ISAP, and PSAP. The BRAP is a living document to be updated at the same time as the CDP, which is also a living document. On the bottom row are the series of assessments described in detail below.

The next section of a BRAP is the pre-specification of benefit-risk assessments for the entire drug development lifecycle. This task is challenging for three reasons. First, the assessments and their assumptions must be prespecified to ensure transparency. Secondly, it is based on a series of decision milestones, whereby the outcome of one assessment impacts the next. Third, its complexity may increase over time. An investigational drug with a clear benefit over a control and no safety concerns may only need a dot-forest plot. The complexity will increase if the treatment difference is small, or a serious but rare adverse event is detected. The FDA Benefit-Risk Framework in Table 1 provides a structure for assessments (see Table 3). It begins with the assessment of the medical condition and current treatment options. Our goal is not to re-evaluate the efficacy and safety of approved treatments; it is to understand for each drug which benefits and risks most impacted past decisions, how they were statistically characterized, and how uncertainty was addressed. Conducting visual assessments [13, 14] facilitates stakeholder discussions of how to assess the investigational drug in the remaining rows of the framework. Identifying different perspectives among stakeholders early in development is an essential part of the decision-making process.

Depending on the need, the assessments have three levels with increasing complexity (see Table 3). The first level is for qualitative treatment-level assessments, which

Table 2 Terminology

Term	Definition
Structured benefit-risk assessment (SBRA)	A transparent decision-making process for determining whether the benefits outweigh the risks. SBRA simultaneously assesses multiple outcomes with different metric scales, time scales, and patient consequences. The decision process integrates evidence with judgment while considering various uncertainties. Heterogeneity among stakeholders, including regulators, physicians, patients, and payers, is also considered.
Uncertainty	Sources of uncertainty include human variability, statistical error, and limitations in the research and clinical trial process that limit the generalizability of results. [10, 11]
Benefit vs. Risk	A benefit is a favorable outcome relative to a control, and a risk is an unfavorable outcome relative to a control.
Treatment-level Assessments	Assessments based on reported or published summary-level statistics for each treatment.
Patient-level assessments	Assessments that use individual patient data to form composite outcomes and assess the statistical association between benefits and risks.
Qualitative assessments	Assessments that assign a qualitative weight or relative importance to each outcome. For example, the assessment orders recurring AEs as least important, serious AEs as important, and fatal AE as extremely important, even though their data are either 0 for did not occur or 1 for did occur. [11]
Quantitative assessments	Assessments that assign a quantitative weight or numeric value (ex., 0 to 1.0) to each outcome so that the outcomes can be combined into an overall score. [11]
Statistical analysis plan (SAP)	Planned analyses for a single study that compares a treatment to a control one endpoint at a time.
BRAP	A plan that first compares multiple benefits to multiple risks, then compares active treatments based on their benefit-risk profiles. More complex assessments require integrating clinical evidence with medical judgment or preference. A BRAP documents assumptions and choices in the decision process but not the results.
Clinical evidence	Objective study results based on a representative sample of study participants.
Value judgment	A subjective value based on expert opinion or personal preference, including levels of utility, risk tolerance, and benefit acceptance, which varies between stakeholders.[11]

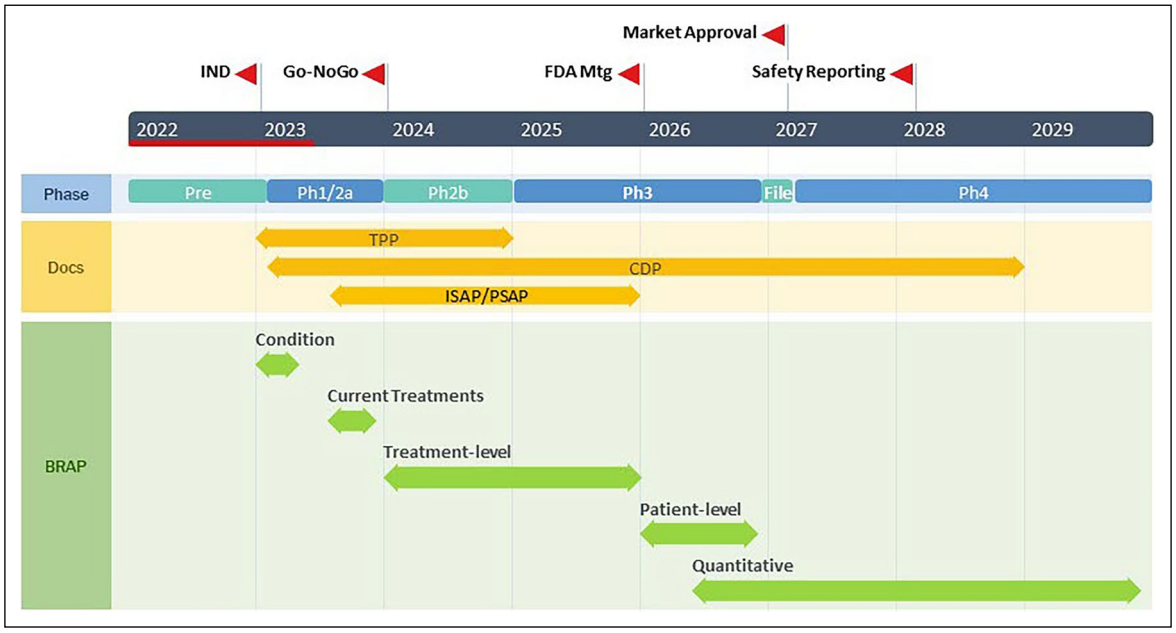

Figure. 1 Gantt Chart of Drug Development Timelines.[12].

assess whether the benefits exceed the risks, e.g., the proportion of patients with a favorable outcome is higher than the proportion with an unfavorable outcome. The second level is for qualitative patient-level assessments, which evaluate the association between benefits and risks, e.g., do the same patients who experience the benefit

Table 3 Mapping FDA framework to BRAP and visualizations

FDA Framework	BRAP Table of Contents	Visualizations
	I. Timelines	
	Decision Milestones	Figure 1. Gannt Chart
	Associated Documents	
Analysis of Condition	II. Analysis of Condition	
	Demographics	Figure 2. Demographics
	Disease history	Figure 3. Disease History
	Comorbidities	Figure 4. Comorbidities
Current Treatment Options	III. Current Treatment Options	
	Key benefits and risks	Figure 5. Value Tree
	Statistically characterized outcomes	Effects table – not shown
	Treatment effects and differences	Figure 6. Dot-Forest Plot
	Acceptance boundaries	Figure 7. Tradeoff Plot
	The relative importance of outcomes	Figure 8. MCDA Stacked Bar chart
		Figure 9. MCDA Benefit-Risk Map
Benefits, Risks and Risk Management	IV. Investigational Drug	
	A. Qualitative treatment-level assessments	
	Repeat assessments for current treatment options	Figures 5, 6, 7, 8, 9 above
	Generalizability	Figure 6. Dot-Forest Plot
		Figure 7. Tradeoff Plot
	B. Qualitative patient-level assessments	
	Association between outcomes	Figure 10. Correlogram
		Figure 11. Scatterplot of Continuous Outcomes
	Asymmetries between outcomes	Figure 12. Stacked Bar Chart – Ordinal Composite Outcomes
		Figure 13. Cumulative Excess Number of Events
	C. Quantitative assessments	
	Weighted tradeoffs	Figure 7. Tradeoff Plot
		Figure 8. MCDA Stacked Bar chart
		Figure 9. MCDA Benefit-Risk Map

also experience the risk? The third level is for quantitative assessments, which integrate clinical evidence with medical judgment or preference, e.g., patients prefer the benefit-risk profile of the new treatment. All three levels use visualization for informed discussions and clear communication of a benefit-risk assessment plan.

We propose one visualization for each assessment in Table 3, resulting in 12 unique figures. Visualizing the medical condition (Figs. 2, 3, 4) and current treatment options (Figs. 5, 6, 7, 8, 9) makes planning assessments for the investigational drug easier. The outcomes and uncertainties most impacting decision-making and the statistics best characterizing the outcomes become evident. A sponsor also gains experience with different types of assessments before applying them to their investigational drug. The BRAP team can schedule assessments as data become available.

Analysis of Condition

To describe the condition and its variability from various angles, the BRAP should address the following:

- Target population and known patient subgroups required for generalizing benefit-risk results
- Aspects of the disease least and most important to patients
- Appropriate timing of assessments based on the natural history of the disease
- Differentiating comorbidities of the disease from adverse effects of a therapy

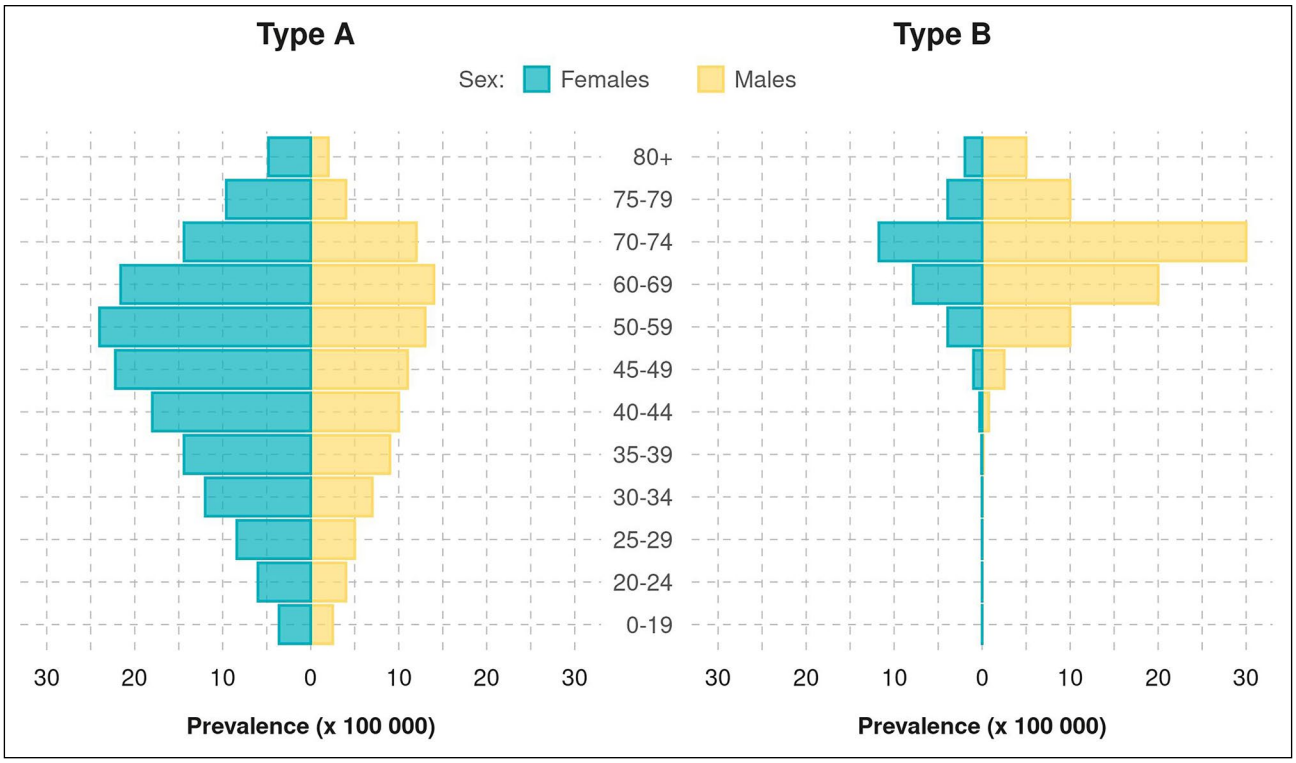

Figure. 2 Demographics [15–17].

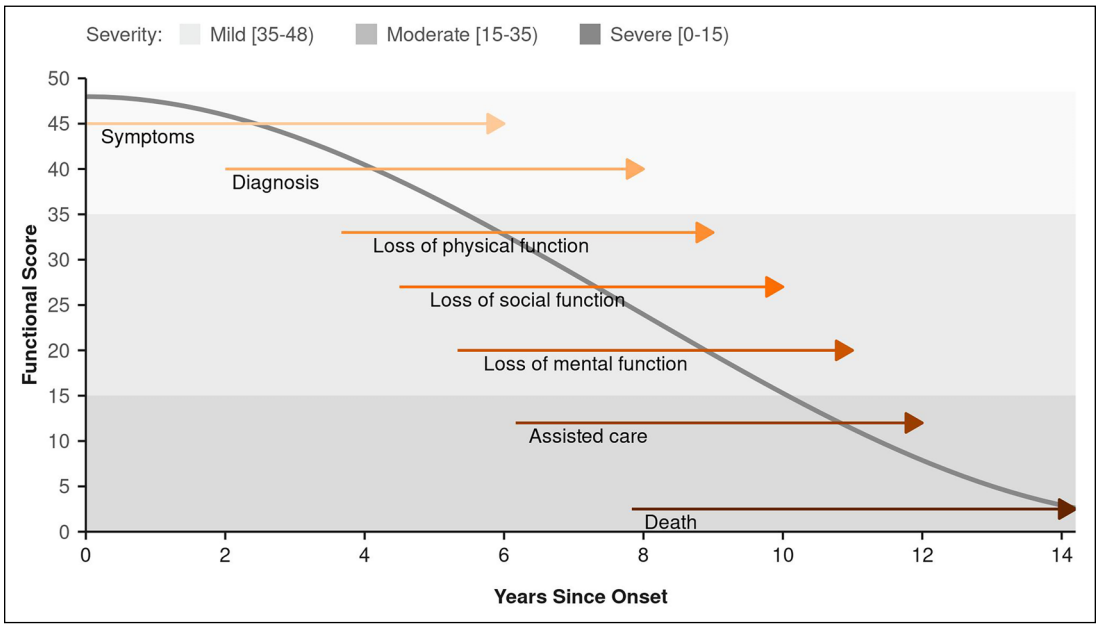

Figure 3 Natural History [15–17].

- Differentiating the benefits of concomitant medications from the benefits of therapy, perhaps using a well-defined estimand.

A description of the medical condition using various means and proportions is not as impactful in planning interventions as using visualization. The three Figs. 2, 3,

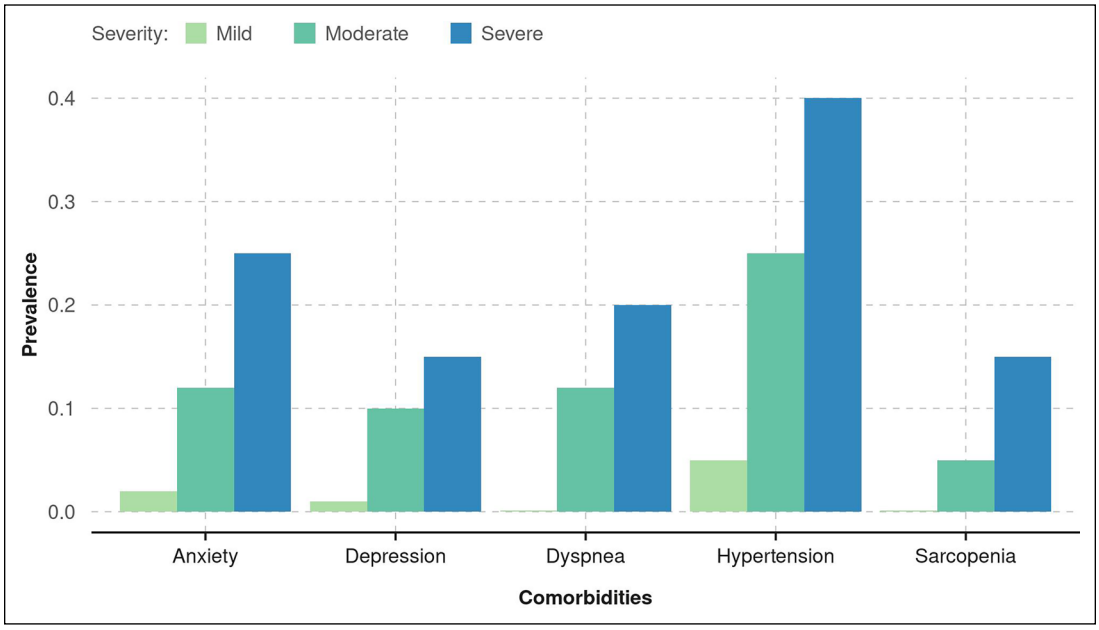

Figure. 4 Comorbidities [15–17].

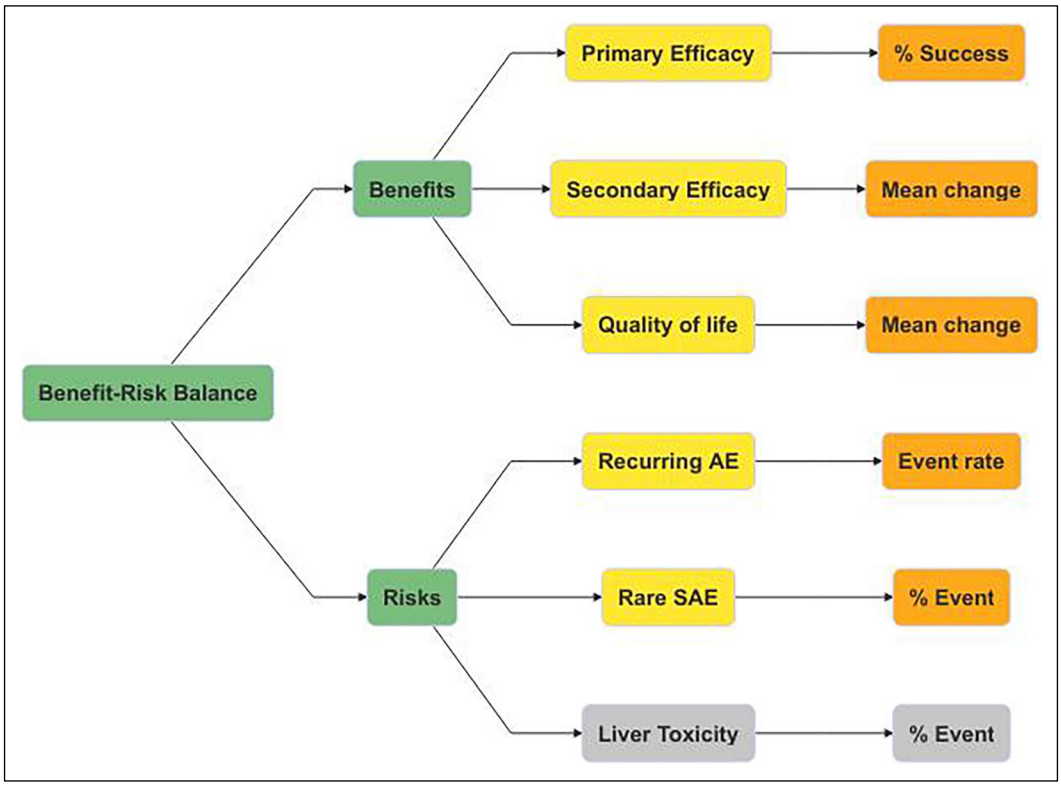

Figure. 5 Value Tree [15, 16, 21].

4 display variability in the hypothetical condition. Figure 2 of epidemiological data displays the different age-sex distributions by disease type, impacting patient engagement and generalization of results across subgroups. Figure 3 of the natural history of the disease shows the association between stage, years since onset, and disease manifestations.

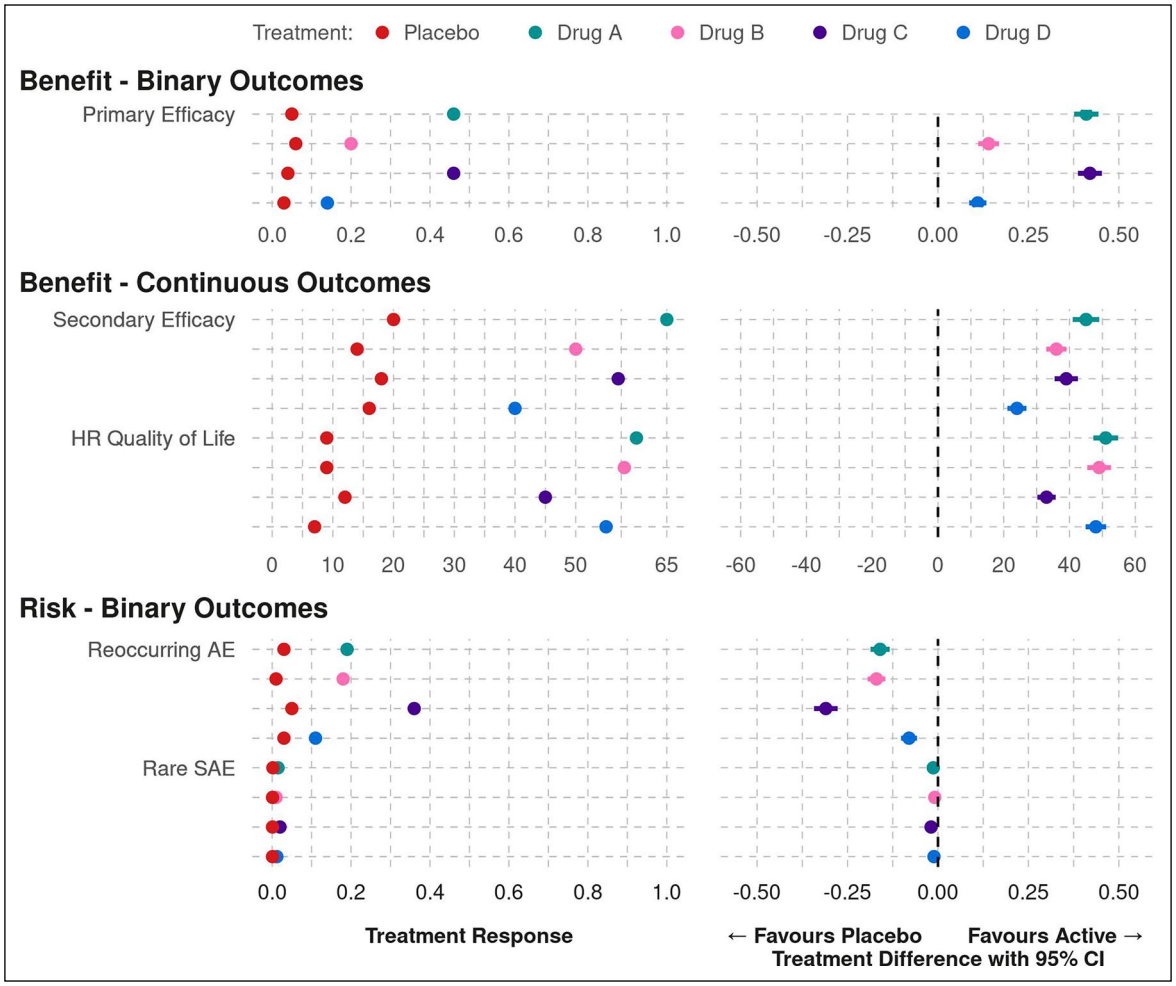

Figure. 6 Dot-Forest Plot [15–17].

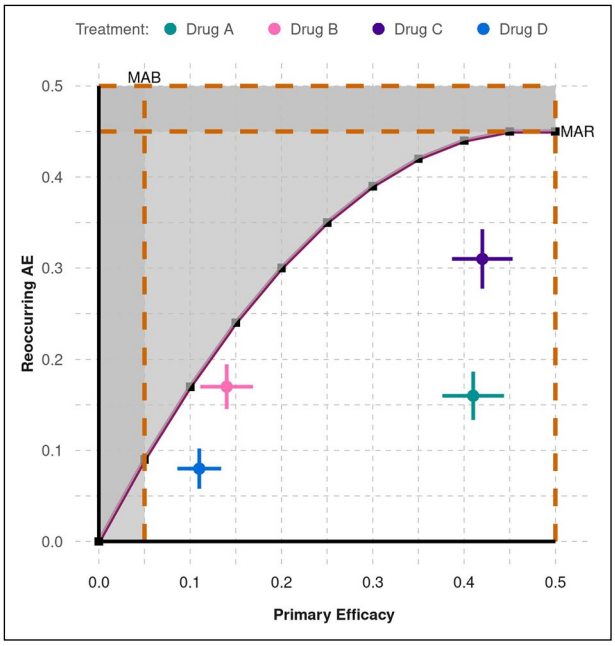

Figure. 7 Tradeoff Plot [15–17].

Health-related quality of life can be characterized by three separate domains or an overall score. Figure 4 displays the top comorbidities by disease stage. This figure helps plan patient preference studies by ordering comorbidities by the level of importance. A plan should also include the obtaining of background rates of comorbidities for determining whether they are reduced or exacerbated by a new therapy.

Current Treatment Options

Assessments of the investigational drug can be well planned by visualizing the published data of current treatment options. Key data sources are the FDA benefit-risk frameworks [5] and labels [18, 19] published on its website for approved treatments. The BRAP can address the unmet need, statistical considerations, and acceptance levels by reviewing several regulatory labels for the same indication. For example, four drug labels might report the same adverse event as a proportion, incidence rate per 1000 patients, incidence rate per 100 patient-years, or exposure-adjusted event rate. The

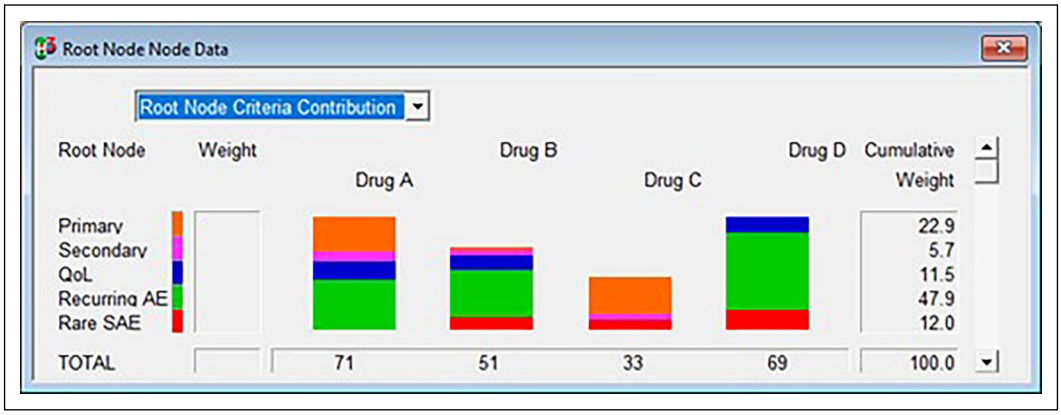

Figure. 8 MCDA Stacked Bar Chart [25].

choice depends on follow-up time, recurrences, etc. Visualizing the benefits and risks together helps select the metric that best characterizes and differentiates the treatments.

To address the second row of the FDA Benefit-Risk framework regarding current treatment options, the BRAP should address the following:

- Outcome selection and characterization using a value tree
- Avoiding bias using a specified tree pruning strategy
- Assessing homogeneity across studies, dosages, and patient subgroups
- Eliciting thresholds for the Maximum Acceptable Risk (MAR), Minimum Acceptable Benefit (MAB), and trade-off threshold curve
- Comparing outcomes having different scales, assessment times, and weights
- Determining when each benefit and risk is acceptable

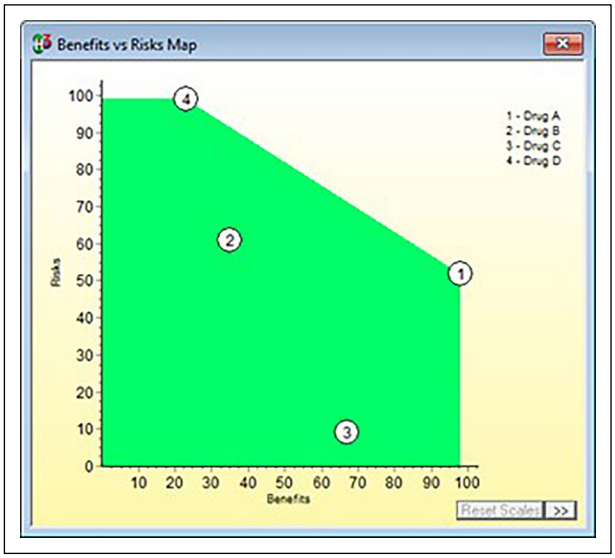

Figure. 9 MCDA Benefit-Risk Map [25].

- Determining the situations where the benefit exceed the risk
- Determining characteristics that a new therapy would need to be preferable to existing therapies
- Sensitivity analyses and assumptions

Continuing with our hypothetical example, we performed a treatment-level qualitative assessment of four current therapies (Drugs A, B, C, and D) and their respective placebo arm. Treatment-level assessments utilize summary statistics published in FDA labels [18] and European Public Assessment Report (EPAR) [20] documents. Qualitative assessments do not combine the outcomes into a single metric, which requires value judgment. The purpose is not to re-evaluate the current treatment options but to understand their statistical characterization and identify an unmet need. In Fig. 5, a value tree displays important outcomes extracted from regulatory labels and patient engagement. In this example, there is a choice of using a mean, mean change from baseline, or proportion of treatment successes for the primary efficacy outcome. There is a choice of proportions, incidence rates, or event rates for adverse events. The BRAP should document the selection of statistics and their assumptions. A strategy should be prespecified for pruning the value tree to outcomes that most impact decision-making. For example, a strategy can first remove highly correlated outcomes, then remove unimportant outcomes, and finally create a composite of several related outcomes. In Fig. 5, liver toxicity is colored gray because it only has a potential association with treatments.

Figure 6 displays the treatment effects on the left and treatment differences on the right. Drugs A and C show greater efficacy on the primary endpoint, but there are tradeoffs with quality of life and reoccurring AEs. The current treatments do not differ in the outcome for the rare SAE, and the BRAP may need to specify a more specific

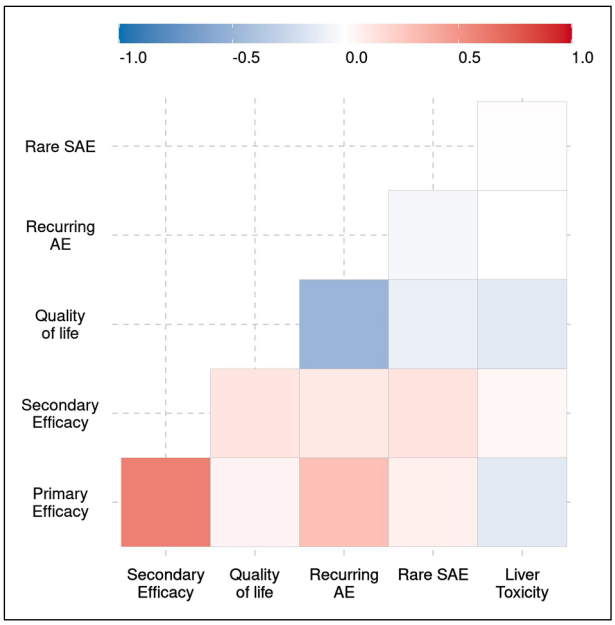

Figure. 10 Correlogram of Binary, Ordered, and Continuous Outcomes [15–17].

SAE for an investigational drug or lengthen the follow-up time. The confidence intervals around the treatment differences display their uncertainty or statistical error. The BRAP should address the potential situation where the difference in benefits is smaller than expected, the difference in risks is larger than expected, or their variability is larger than expected. Figures can explore homogeneity between subgroups if data are available.

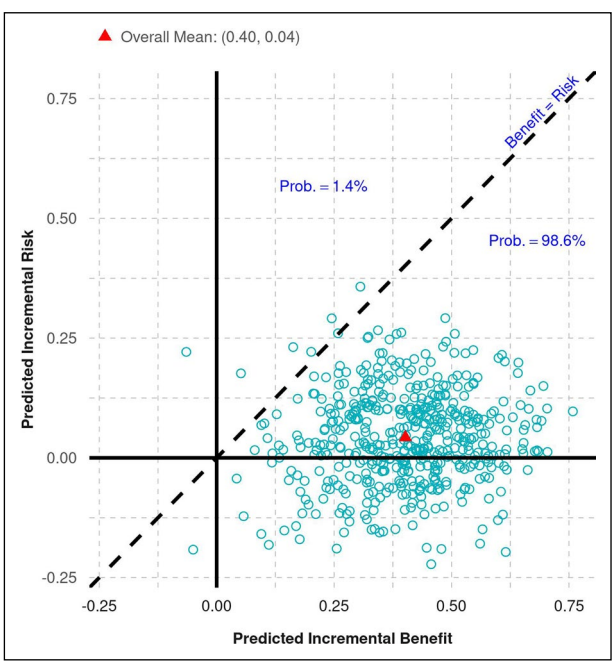

Figure. 11 Scatter Plot of Continuous Outcomes [15–17].

Figure 7 is a tradeoff plot showing the tradeoffs in using one drug over another. The crosses are the same points and confidence interval shown in Fig. 6, but the confidence intervals are displayed vertically for risk and horizontally for benefit. While Fig. 6 displays all outcomes for one drug, Fig. 7 displays a single benefit-risk pair to compare all drugs. Three acceptance boundaries, MAR and MAB (dashed lines), and the tradeoff threshold curve (solid curved line) are elicited from stakeholders, including physicians and patients [22]. Multiple thresholds curves can be displayed to represent different stakeholders. Any point above the horizontal MAR and to the left of the vertical MAB falls in the unacceptance region. Clinically meaningful differences specified in the TPP and SAP may be a source for the MAR and MAB. Patients are willing to accept a higher risk for a higher benefit within these two thresholds, displayed as the curved tradeoff curve. The tradeoff technique [23] elicits valid values. Below this curve is the white acceptance region. For planning purposes, assume that all marketed drugs fall in this white region (i.e., benefits exceed the risks). The option closest to the lower right corner is the most favorable one for this benefit-risk pair. In the BRAP, a sponsor can plan a physician or patient preference study to quantify these thresholds better.

Figure 7 shows Drug A to be more favorable than Drugs B and C. There is a tradeoff between Drugs A and D. Severe patients may prefer Drug A for its higher benefit, and mild patients may choose Drug D for its lower risk. Fig. 7 can display other combinations of one benefit and one risk, as well as for different patient subgroups, dose levels, and studies, in support of the generalizability of the investigational drug. Ideally, the investigational drug will appear in an area of unmet need in the acceptance region.

The following two figures most directly address the challenges of comparing multiple drugs with multiple outcomes having different measurement scales and weights. Figures 8 and 9 use a quantitative approach called Multi-Criteria Decision Analysis (MCDA) [24]. It uses clinical results and quantitatively assessed weights. Using decision software, such as HiView3 [25], users can integrate value judgment and clinical evidence. It calculates a total preference value by mapping each treatment difference for benefits and risks to a standard scale (0 to 100 units), using a value function, and assigning it a weight relative to the other outcomes using a method such as swing-weighting [26].

For this paper, MCDA adds transparency to the planned decision process by explicitly documenting the weighting technique. During drug development, MCDA can be used to assess the investigational drug. While MCDA utilizes informed but subjective value functions and weights, it is more transparent and quantifiable than qualitative (unweighted) assessments. The sponsor can outsource its MCDA and preference studies to a contract research

organization (CRO) specializing in value functions and weights. Figure 8 lists the three benefits and two risks on the left and displays their respective weighted values as color bands in the stacked bar chart. The number of benefits to risks should not tip the balance in favor of a drug. Thicker bands and higher total values indicate a stronger preference. Secondary efficacy has little weight due to its correlation with primary efficacy, and rare SAE carries little weight due to its narrow range of treatment differences. Other risks better differentiate the drugs. Stakeholders will marginally prefer Drug A to Drug D and much prefer Drug A to Drugs B, and C. Drug A performed better on primary efficacy, and Drug D performed better on recurring adverse events.

Figure 9 displays the total scores for benefits versus risk. The figure appears similar to the tradeoff plot above but with significant differences. Higher is better on each axis. Rather than display a single benefit-risk pair, it encompasses all outcomes at the same time. The best option is on the green boundary closest to the upper right corner. There is a tradeoff between Drugs A and D, with Drug A having a more favorable benefit and Drug D having a more favorable risk. Drugs B and C, inside the green region, are dominated by Drugs A and D. Stakeholders should use the MCDA assessment of current treatments to select outcomes and weights for the investigational drug. A sensitivity analysis will show how different stakeholder preference weights can alter the total scores and tip the decisions.

Investigational Drug

Repeat the treatment-level assessments of current treatment options visualized in Figs. 5, 6, 7, 8, 9 for the investigational drug. To ensure generalizability, replicate the dot-forest and tradeoff plots for different doses, studies, and patient subgroups. Patient-level and quantitative assessments can be planned if further questions or uncertainties remain. Patient-level data are required to address the following:

- Association between benefit and risk outcomes
- Benefit-risk profile over time
- Variation among patients
- Identifying patients benefiting most/least from the investigational drug.

One question that cannot be answered with treatment-level statistics is whether the patients who experience the

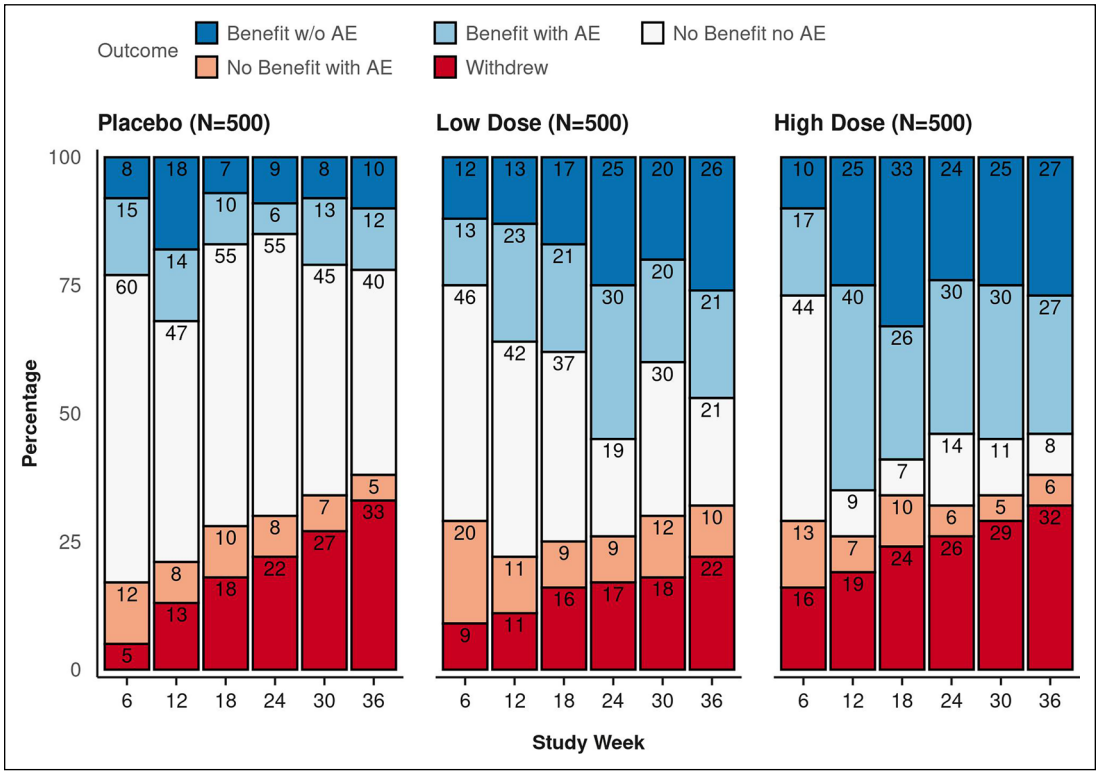

Figure. 12 Stacked Bar Chart of Ordinal Composite Outcomes [15–17].

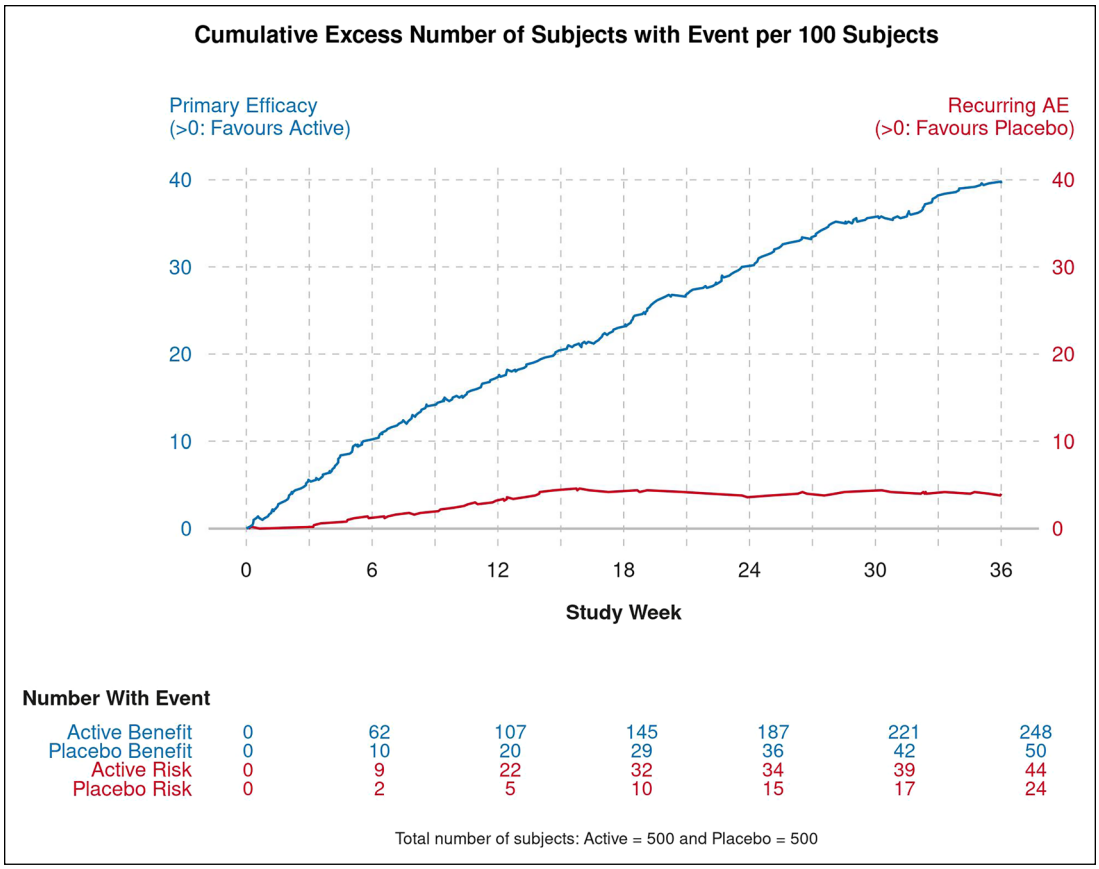

Figure. 13 Cumulative Excess Events [14–16].

benefit also share the risk. Only patient-level statistics, such as correlation coefficients, can show whether risk increases with benefit. The correlogram of binary, ordinal, and continuous variables also assesses whether two benefits or two risks are so highly correlated that only one is needed to avoid double counting. The correlogram in Fig. 10 shows an expected high association between the primary and secondary efficacy outcomes. There may be a concern with double counting efficacy. As expected, a negative association exists between outcomes for quality-of-life and recurring AE. Figure 11 displays the predictive probabilities of experiencing a benefit and a risk with 98.6% of patients in the acceptance region (i.e., below the threshold) [27, 28]. The probabilities are estimated using logistic regression models with demographic and baseline covariates, but separately for one benefit and one risk. Besides displaying patient variability, it indicates which patients are most likely to benefit from the investigational drug. For patients in the upper left area, the risk exceeds the benefit. For patients in the lower right area, the benefit exceeds the risk. The diagonal threshold depicts the benefit and risk having equal weight. As the slope increases, the benefit receives more weight than the risk, meaning patients are willing to accept much more risk in exchange for more benefit. Figure 12 displays a benefit-risk composite of binary outcomes, ranging from "Benefit with no AE" to "No Benefit with AE" [29]. As subjects withdrew from the study, they continued to be counted in the "Withdrew" category at subsequent visits. As expected, the placebo control group has more patients with "No Benefit No AE". The proportion of patients with benefits is highest in the high-dose group. The figure also displays whether the overall benefit-risk profile is sustainable over clinical visits.

Figure 13 assesses whether the binary events for a benefit and risk occur concurrently and whether the profile is sustainable. The benefit is an increase in primary efficacy between the investigational treatment and its placebo control, and the risk is an increase in recurring adverse events. The difference in cumulative incidence rates per 100 subjects (i.e., active minus placebo) is calculated at each time point. The upward trends mean both the benefit and risk are higher for the investigational drug than placebo, but not at the same rate. Ideally, the benefit would increase with drug exposure while the risk plateaus.

Summary

The BRAP is a living document with two sections, one for timelines and the other for pre-specifying the assessments. Assessments of the medical condition and current treatment options provide insight into assessing the investigational drug. Visualization facilitates stakeholder discussions on how to characterize the outcomes and minimize uncertainties in decision-making. Stakeholders can identify the need for new patient-centered outcomes, preference data, and more complex assessments. Ultimately, pre-specification of required assessments and data will impact study designs, thereby strengthening the evidence and reducing uncertainty in decision-making internally by sponsors and externally by regulatory agencies.

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Author Contributions
A small internal working group developed this benefit-risk visualization methodology under the auspices of the Statistical Sciences & Innovation Safety Standards team. All authors drafted the initial manuscript, revised and approved the final manuscript draft for submission, and agreed to be accountable for all aspects of the work.

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Data Availability
No data is presented in this manuscript; however, further information on any of the content can be requested from the corresponding author.

Declarations

Conflict of interest
All authors are employees of UCB Pharma and may hold UCB stock or stock options.

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